

Initial Date: 9/2004

Revised Date: 03/24/2023

Section: 10-5

Cyanide Exposure

Purpose: This Protocol is intended for EMS personnel at all levels to assess and treat patients exposed to cyanide. Additionally, the protocol allows trained and authorized paramedics to administer antidotes when available.

NOTE: A single medical control order in a mass casualty incident may be applied to all symptomatic patients.

Definitions: For the purposes of this protocol Cyanokit (brand name) refers to **Hydroxocobalamin**

Medications in this protocol are not required to be carried on EMS vehicles and may be available through special response units.

Chemical Agent

1. Agents of Concern: Cyanide
 - a. Hydrogen Cyanide
 - b. Potassium/Sodium Cyanide
 - c. Cyanogen Chloride
2. Detection: The presence of these agents can be detected through specialized environmental monitoring equipment available to hazardous materials response teams.
3. Modes of Exposure
 - a. Inhalation (including smoke inhalation)
 - b. Ingestion
 - c. Skin absorption unlikely
4. Alert receiving hospital ASAP to prepare additional antidotes

Assessment

1. Hypotension
2. Shortness of breath
 - a. Possibly accompanied by chest pain
 - b. Generally, not associated with cyanosis
 - c. Pulse oximetry levels usually normal
 - d. Usually associated with increased respiratory rate and depth
 - e. Potential for rapid respiratory arrest
3. Confusion, decreased level of consciousness, coma
4. Seizures
5. Headache, dizziness, vertigo (sense of things spinning)
6. Pupils may be normal; dilation is a late sign

Indications for Antidote use in patient with suspected cyanide poisoning:

1. Cardiac or Respiratory Arrest
2. Hypotension SBP<90 mm Hg
3. GCS ≤ 9

Personal Protection

1. Be Alert for secondary device in potential terrorist incident

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2. Personal Protective Equipment (PPE) as directed by Incident Commander.
3. Assure EMS personnel are operating outside of Hot and Warm Zones, unless appropriately trained and in proper PPE.
4. Avoid contact with vomit if ingestion suspected – off gassing possible
5. Decontamination of victims usually not indicated unless additional unknown chemical(s) suspected

Patient Management (in Cold zone)

1. Administer oxygen 10-15 LPM via non-rebreather mask and support ventilation as needed. Per **Oxygen Administration-Procedure Protocol and/or Airway Management-Procedure Protocol**

- a. Note: Patients in respiratory arrest (i.e., not breathing but still having a pulse) have been found to respond to antidote therapy and should receive positive pressure ventilation when operationally feasible.
- b. This is in contrast to most triage systems that would categorize non-breathing patients as non-survivable.



2. Establish vascular access. Refer to **Vascular Access & IV Fluid Therapy-Procedure Protocol**



3. Administer antidote:

- a. **Cyanokit®** (5g. adult IV/IO; 70 mg/kg pediatric IV/IO) per **Hydroxocobalamin (Cyanokit®)-Medication Protocol** (preferred, per MCA Selection)

Cyanokit® Included?



Yes



No

- b. Each vial of **Cyanokit®** for injection is to be reconstituted with diluent (not provided with **Cyanokit®**) using the supplied sterile transfer spike.
 - i. The recommended diluent is **0.9% Sodium Chloride** injection (0.9%NaCl).
 - ii. The line on each vial label represents the volume of diluent. Following the addition of diluent to the lyophilized powder, each vial should be repeatedly inverted or rocked, not shaken, for at least 60 seconds for the 5g bottles, 30 seconds for the 2.5g bottles prior to infusion.
 - iii. **Cyanokit®** solutions should be visually inspected for particulate matter and color prior to administration.
 - iv. If the reconstituted solution is not dark red or if particulate matter is seen after the solution has been appropriately mixed, the solution should not be administered to the patient and should be discarded.
 - v. There are a number of drugs and blood products that are incompatible with **Cyanokit®**, thus **Cyanokit®** requires a separate intravenous line for administration.
 - vi. Depending upon the severity of the poisoning and the clinical response, a second dose of 5 g may be administered by IV/IO infusion for a total dose of 10g in adults. The rate of infusion for the second dose may range from 15 minutes (for patients in extremis) to two hours, as clinically indicated.



Contact medical control for second dose instructions for pediatric patients.

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Cyanokit® Administration for Suspected Cyanide Poisoning (including serious smoke inhalation)

Weight	Age	Cyanokit® Dose ¹ (~70 mg/kg +/-) IV/IO	Cyanokit® Volume to Administer ² IV/IO
3-5 kg (6-11 lbs)	0-2 months	250 mg	10 mL ³
6-7 kg (13-16 lbs)	3-6 months	500 mg	20 mL ³
8-9 kg (17-20 lbs)	7-10 months	625 mg	25 mL ³
10-11 (21-25 lbs)	11-18 months	750 mg	30 mL ³
12-14 kg (26-31 lbs)	19-35 months	900 mg	36 mL ³
15-18 kg (32-40 lbs)	3-4 years	1100 mg	44 mL ³
19-23 kg (41-51)	5-6 years	1500 mg	60 mL ³
24-29 kg (52-64)	7-9 years	1750 mg	70 mL ³
30-36 kg (65-79 lbs)	10-14 years	2500 mg	100 mL ⁴ (1/2 bottle)
Adult 37 40 kg (80 88 lbs)	>14 years	3000 mg	120 mL ⁴
Adult 41 49kg (89 108 lbs)	>14 years	3500 mg	140 mL ⁴
Adult > or 50 kg (> or 109 lbs)	>14 years	5000 mg	200 mL ⁴ (full bottle)

¹The safety and efficacy in pediatrics has not been established, ²Administer slowly over 15 minutes.

³Push slowly over 15 minutes, ⁴Infuse over 15 minutes

4. Cardiac monitoring

5. Special Considerations for Smoke Inhalation

- Smoke inhalation victims may have cyanide poisoning along with burns, trauma, and exposure to other toxic substances making a diagnosis of cyanide poisoning particularly difficult.
- Prior to administration of **Cyanokit®**, smoke inhalation victims should be assessed for the following:
 - Exposure to fire or smoke in an enclosed area
 - Presence of soot around the mouth, nose or oropharynx
 - Altered mental status
- The **Cyanokit®** should be considered for all serious smoke inhalation victims (including cardiac arrest).

Medication Protocols

Hydroxocobalamin (Cyanokit®)